MAS · Complex Numbers — Question Patterns and Solution Strategies

Basis: marking keys + exam reports for a total of 43 past complex-number questions from 2016–2025. Original questions in `topics/01_Complex_Numbers.pdf`. Numerical examples verified with sympy. Also cites 2 questions from the 2016 sample paper (`2016S-F1`, `2016S-A18`; original paper in `papers/2016Sample_*`). Review status: has been through one round of independent review and revised according to the feedback (see `Review_log.md`).

Examiner reminders

- Use polar form when you should (2016 report: those who didn't convert $(\sqrt{3} - i)^5$ to polar form lost a lot of marks). Always use polar form for powers and roots; for multiplication and division it depends on the form (e.g. $(2+i)/(1-i)^2$ is faster using the conjugate, 2016F-Q2(a)).
- Arguments in the third quadrant (the 2020 report specifically names 2020A-Q15(a): a third-quadrant complex number was not converted to polar form correctly): for example, the principal argument of $-1 - i$ is $-3\pi/4$, not $\pi/4$, and $5\pi/4$ is not allowed either (outside the principal range). Sketch first to confirm the quadrant.
- Vector interpretation of complex numbers (2018, 2021, 2025 reports: $z + r$, the argument of $-z$, regular hexagon): do $z + r$, $z - k$ (e.g. 2023A-Q12) and $-z$ by drawing them as vectors (rhombus, isosceles triangle, regular hexagon).
- Basics of expanding brackets (2021, 2023, 2024 reports): $(z - (2+4i))(z - (2-4i)) = z^2 - 4z + 20$ is often miscalculated; for $z^2 - 2z + 3 = 0$ use completing the square or the quadratic formula, and do not split it into $(z - 3)(z + 1)$.
- Know your exact trig values (2019, 2021, 2023 reports).
- "Show that $(z - 2i)$ is a factor": after substituting, you must write out the calculation for each term; the key says "provides evidence, i.e. not just the statement $P(2i) = 0$".

Pattern 1: Operations in Cartesian form and converting between forms (mostly in CF)

Approach

- Division: multiply the numerator and denominator by the conjugate of the denominator.
- Converting to polar form: $r = \sqrt{a^2 + b^2}$; determine θ from the quadrant. Example: given $z = 3 + bi$ and $\tan\theta = 2 \Rightarrow b = 6$, $r = \sqrt{45} = 3\sqrt{5}$ (2024F-Q1).
- Higher powers: $(\sqrt{3} - i)^5 = (2\,\mathrm{cis}(-\pi/6))^5 = 32\,\mathrm{cis}(-5\pi/6) = -16\sqrt{3} - 16i$. The key also accepts binomial expansion, but it takes several more steps.

Mark-earning points: ✓ modulus ✓ argument ✓ use De Moivre ✓ convert to $a + bi$. Past questions: 2016F-Q2, 2017A-Q10(a), 2019A-Q12(a)(b), 2021A-Q11(a)(b), 2024F-Q1.

Pattern 2: Solving $z^n = w$ (finding nth roots) ★ tested every year

Standard procedure

1. Write w as R cis φ (1 mark for modulus, 1 mark for argument).
2. z = R^{1/n} cis((φ + 2kπ)/n), k = 0, 1, …, n−1 (1 mark for writing the general form).
3. Within the range required by the question (commonly $-\pi < \theta \le \pi$), list all n roots; adjacent roots differ in argument by $2\pi/n$.
4. Example: $z^4 = -16i = 16\,\mathrm{cis}(-\pi/2) \Rightarrow z = 2\,\mathrm{cis}(-5\pi/8), 2\,\mathrm{cis}(-\pi/8), 2\,\mathrm{cis}(3\pi/8), 2\,\mathrm{cis}(7\pi/8)$. $z^4 = 2 - 2\sqrt{3}i = 4\,\mathrm{cis}(-\pi/3) \Rightarrow \sqrt{2}\,\mathrm{cis}(-7\pi/12), \sqrt{2}\,\mathrm{cis}(-\pi/12), \sqrt{2}\,\mathrm{cis}(5\pi/12), \sqrt{2}\,\mathrm{cis}(11\pi/12)$.

Variations

- Given the diagram of one root, work backwards to find k (2025F-Q1): $w_2^3 = ki$.

- Number of roots in the first quadrant ⇒ find the possible values of n; you must give a reason (2017A-Q19(b)).
- Roots of $z^n = 1$: on the unit circle, evenly spaced, separated by $2\pi/n$ (2024A-Q9(c) explanation worth 2 marks: circle of radius 1, equal spacing).
- Product of the roots (2019F-Q9): the product of all roots = (−1)^{n+1}; it is 1 when n is odd and −1 when n is even.
- Common roots: $z^{43} = 1$ and $z^{47} = 1$ have only z = 1 as a common root, because 43 and 47 are coprime (2021F-Q7).
- z^n a positive real number (2022F-Q6(c)): n·arg z must be an integer multiple of 2π. $z_1 = 4\,\mathrm{cis}(5\pi/6)$ ⇒ 5n/6 = 2k ⇒ smallest n = 12, $r = 4^{12} = 2^{24}$.

Past questions: 2016A-Q14, 2017A-Q19, 2018F-Q7(a), 2020A-Q13, 2022A-Q18(b), 2022F-Q6, 2023F-Q6, 2024A-Q9, 2025F-Q1.

Pattern 3: Complex roots of polynomials (factor theorem + conjugate root theorem)

Standard procedure

1. "Show that (z − α) is a factor": calculate P(α), writing out each term, and get 0.
2. A polynomial with real coefficients ⇒ the conjugate $\bar{\alpha}$ is also a root ⇒ $(z - \alpha)(z - \bar{\alpha}) = z^2 - 2\mathrm{Re}(\alpha)z + |\alpha|^2$ is a real quadratic factor.
3. Polynomial division or equating coefficients to find the remaining factor(s).
4. Solve the remaining quadratic (completing the square or the quadratic formula).

Example (2021F-Q6): P(2 + 4i) = 0 ⇒ factor $z^2 - 4z + 20$ ⇒ $P = (z^2 - 4z + 20)(z^2 - 2z + 3)$ ⇒ $z = 2 \pm 4i, 1 \pm \sqrt{2}\,i$.

Working backwards to find coefficients (2023F-Q2): given roots 1 + i and 2 + √3 i ⇒ the conjugate roots are also roots ⇒ $Q(z) = (z^2 - 2z + 2)(z^2 - 4z + 7)$ ⇒ use the constant term 14 to deduce the real root z_0, then expand and equate coefficients.

Past questions: 2016F-Q3, 2016S-F1, 2017F-Q2, 2018F-Q7(b)(c), 2019F-Q2, 2021F-Q6, 2022A-Q13(a)(b), 2023F-Q2, 2024F-Q7.

Pattern 4: Geometric meaning of multiplication (rotation + dilation)

Recognition: "Describe the geometric transformation performed by successive multiplication by z".

Template (2 marks): "Multiplication by $w = r\,\mathrm{cis}\,\theta$ dilates the modulus by a factor r and rotates the vector anticlockwise by θ (or clockwise by |θ| if θ < 0) about the origin." 1 mark for each of the two elements; multiplying by z^{-1} takes the reciprocal of the factor and reverses the direction of rotation (2023A-Q10(d)).

Extensions

- Find all n for which zw^n lies in the second quadrant: each multiplication by w rotates 30°, so list the n with 90° < argument < 180°. When |w| = 1 it returns to its original position every 12 multiplications, so the answer must be complete: n = 3 + 12k, 4 + 12k, 5 + 12k (k a non-negative integer); writing only 3, 4, 5 does not earn the "all the possible values" mark (2025A-Q19(b)).
- Read the modulus and argument of z from the diagram, then write the polar form (2019A-Q12(b)).

Past questions: 2017A-Q10, 2019A-Q12, 2023A-Q10, 2025A-Q19.

Pattern 5: Loci — sketching from an equation

Equation or inequality	Graph
$\|z - a\| = r$	Circle with centre a and radius r
$\|z - a\| = \|z - b\|$	Perpendicular bisector of the line segment joining a and b

$\arg(z - a) = \theta$	Ray starting at a (the endpoint a is not on the locus; draw an open circle)
$\lvert z - a\rvert + \lvert z - b\rvert = k$	An ellipse when $k > \lvert a - b\rvert$; degenerates to the line segment ab when $k = \lvert a - b\rvert$ (2022A-Q9(a))
$\lvert z + 2\rvert = \lvert z - i\rvert + \sqrt{5}$	The difference equals the distance between the two points ⇒ a ray starting at i and extending in the direction $-2 \to i$; the endpoint i is on the locus (solid dot), the opposite of an Arg-type ray (the 2020A-Q10(b) key gives 1 mark specifically for this)
$(z + i)\cdot\overline{(z + i)} = 2$	$\lvert z + i\rvert^2 = 2$ ⇒ centre $-i$, radius $\sqrt{2}$ (2022A-Q9(b))
$z - \bar{z} \le 6i$	$2i\cdot\mathrm{Im}(z) \le 6i$ ⇒ $\mathrm{Im}(z) \le 3$ (2024A-Q10(b))

Mark-earning points for the sketch: ✓ key points or centre ✓ position of the boundary ✓ solid/dashed boundary ✓ correct shaded region.

Past questions: 2016A-Q10(a)(b), 2016S-A18, 2020A-Q10(b), 2022A-Q9(a)(b), 2024A-Q10(b), 2025A-Q10(b).

Pattern 6: Loci — writing an equation from a graph

Approach: identify the shape first, then write the equation in terms of z (you cannot use x, y): circle ⇒ $\lvert z - \text{centre}\rvert \le r$; ray ⇒ $\arg(z - \text{endpoint}) = \theta$; sector ⇒ use two Arg inequalities plus one modulus inequality. Work out the angle at the endpoint exactly using trigonometry (2020A-Q10(a)).

Past questions: 2018A-Q11(a)(b), 2019A-Q10(a), 2020A-Q10(a), 2021A-Q11(d), 2022A-Q9(c), 2025A-Q10(a).

Pattern 7: Maximum and minimum values on a locus

Approach: sketch → find the geometric position where the maximum/minimum occurs → calculate using Pythagoras or trigonometry.

- Maximum distance from a point on the circle to c = $\lvert\text{centre} - c\rvert + r$, minimum distance = $\lvert\text{centre} - c\rvert - r$ (when c is outside the circle; when c is inside, it is $r - \lvert\text{centre} - c\rvert$) (2016A-Q10(c), 2024A-Q10(a)(ii)).
- Maximum argument of a point on the circle: occurs at the tangent position. The θ from $\sin\theta = r/d$ is only "the angle between the tangent and the line O→centre"; you then have to add or subtract the argument of the centre, and watch the argument range specified in the question (2018A-Q11(c): $0 \le \arg z < 2\pi$, answer $2\pi - \alpha \approx 5.64$).
- Minimum distance from a point on the ray to i: drop a perpendicular (2019A-Q10(b)).

Mark-earning points: ✓ indicate on the diagram where the maximum/minimum occurs ✓ write the expression ✓ exact value.

Pattern 8: Finding modulus and argument using vectors or geometry (appears in both CF and CA)

Approach: Draw z + w as a parallelogram and z − k as a triangle, then use:

- equal moduli of two vectors ⇒ rhombus ⇒ diagonals bisect the angle: $\arg(z + r) = \theta/2$ (2018F-Q3(b)(ii));
- isosceles triangle, inscribed angle theorem, sine rule or cosine rule (2017F-Q6, 2021F-Q1(b), 2023A-Q12);
- regular hexagon: when centred at the origin, adjacent vertices differ in argument by $\pi/3$ and have equal moduli. For 2025F-Q7, the vertex z_0 is at the origin, and the hexagon is not centred at the origin: the side vectors rotate successively by $\pi/3$; using the cosine rule gives $\lvert z_2\rvert = \lvert z_4\rvert = \sqrt{3}r$, $\lvert z_3\rvert = 2r$, so $\lvert z_1z_2z_3z_4z_5\rvert = r\cdot\sqrt{3}r\cdot 2r\cdot\sqrt{3}r\cdot r = 6r^5$; $z_3 = 2r\,\mathrm{cis}(\pi/3)$, $\arg(z_5 - z_3) = -5\pi/6$ (the 2025 report specifically highlights the hexagon geometry in this question).

Past questions: 2017F-Q1, 2017F-Q6, 2018F-Q3, 2020A-Q11, 2020A-Q15, 2021F-Q1, 2023A-Q12, 2025F-Q7.

Pattern 9: Proof and technique questions

- Identities: use conj(z) = r cis(−θ), together with cos being even and sin being odd (2018A-Q10).
- Summation by grouping: $n \cdot i^n$ is not itself periodic, but every 4 consecutive terms (terms 4m+1 to 4m+4) sum to 2 − 2i, with 505 groups in total ⇒ $\Sigma_{n=1}^{2020} n\, i^n = 505 \times (2 - 2i) = 1010 - 1010i = 1010\sqrt{2}\,\text{cis}(-\pi/4)$ (2020F-Q8).
- Solving real simultaneous equations using complex numbers: $(a + bi)^3 = (a^3 - 3ab^2) + i(3a^2b - b^3) = 14 - 2\sqrt{5}\,i$ ⇒ taking the modulus: $(a^2 + b^2)^3 = 196 + 20 = 216$ ⇒ $a^2 + b^2 = 6$ (2023F-Q8).
- Using complex multiplication to prove the arctangent sum (2025A-Q19(c)).